NOTRE DAME UNIVERSITY BANGLADESH

CSE LAB Report

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Course Title: Object Oriented Programming Lab

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Lab Task Topic: Problems of Java Language using functions.

Submitted to:

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Problem No. 01: Grade Point using Switch case.

```java
//Grade Point CGPA
import java.util.Scanner;

public class Code1_GradeCGPA {
    public static int TotalMark(int mat, int phy, int chem, int
eng, int cse) {
        int total = mat + phy + chem + eng + cse;
        return total;
    }

    public static int Average(int total_mark) {
        int average = total_mark / 5;
        return average;
    }

    public static void main(String[] args) {
        System.out.println("\n");
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your mark in Mathematics:");
        int mat = input.nextInt();

        System.out.print("Enter your mark in Physics:");
        int phy = input.nextInt();

        System.out.print("Enter your mark in Chemistry:");
        int chem = input.nextInt();

        System.out.print("Enter your mark in English:");
        int eng = input.nextInt();

        System.out.print("Enter your mark in CSE:");
        int cse = input.nextInt();

        int total_mark = TotalMark(mat, phy, chem, eng, cse);

        int average = Average(total_mark);
```

```
System.out.println("\nYour Average Mark is: " + average);
   int mark = average;
   String grade = null;
   double cgpa = 0.00;

   switch ((mark > 79) ? 1 : 0) {
       case 1:
           grade = "A+";
           cgpa = 4.00;
           break;
       case 0:
           switch ((80 > mark && mark > 74) ? 1 : 0) {
               case 1:
                   grade = "A";
                   cgpa = 3.75;
                   break;
               case 0:
                  switch ((75 > mark && mark > 69) ? 1 : 0) {
                           case 1:
                               grade = "A-";
                               cgpa = 3.50;
                               break;
                           case 0: {
                switch ((70 > mark && mark > 64) ? 1 : 0) {
                                case 1:
                                    grade = "B+";
                                    cgpa = 3.25;
                                    break;
                                case 0:
                  switch ((65 > mark && mark > 59) ? 1 : 0) {
                                     case 1:
                                            grade = "B";
                                            cgpa = 3.00;
                                        break;
                                       case 0:
                  switch ((60 > mark && mark > 54) ? 1 : 0)
{                                          case 1:
                                          grade = "B-";
                                           cgpa = 2.75;
                                              break;
                                           case 0:
                switch ((55 > mark && mark > 49) ? 1 : 0) {
```

```
case 1:
grade = "C+";
cgpa = 2.50;
break;
                    switch ((50 > mark && mark > 44) ? 1 : 0) {
    case 1:
        grade = "D";
        cgpa = 2.00;
        break;
    case 0:
        grade = "F";
        cgpa = 0.00;
        break;
}

break;
                                                        }
                                                        break;
                                                }
                                                break;
                                        }
                                        break;
                                }
                                break;
```

```
                                }

                            }

                    }
            }
System.out.println("Your Grade is : " + grade + "\nYour CGPA is
: " + cgpa);
    }

}
```

Problem No. 02: Show current time.

```
//show current time in windows
import javax.swing.JOptionPane;

public class Code2_ShowCurrentTime {
    // Display results
public static void Windows(long localtime, long currentMinute,
long currentSecond) {
JOptionPane.showMessageDialog(null, "Current time is " +
localtime + " : "+ currentMinute + " : " + currentSecond + "
GMT");
System.out.println("Current time is " + localtime + " :"
+ currentMinute + " : " + currentSecond + " GMT");
    }

 public static void main(String[] args) {
  // Obtain the total milliseconds since midnight, Jan 1, 1970
        long totalMilliseconds = System.currentTimeMillis();
        // Obtain the total seconds since midnight, Jan 1, 1970
        long totalSeconds = totalMilliseconds / 1000;
        // Compute the current second in the minute in the hour
        long currentSecond = totalSeconds % 60;
        // Obtain the total minutes
        long totalMinutes = totalSeconds / 60;
        // Compute the current minute in the hour
        long currentMinute = totalMinutes % 60;
        // Obtain the total hours
        long totalHours = totalMinutes / 60;
        // Compute the current hour
        long currentHour = totalHours % 24;
        long localtime = currentHour;

        Windows(localtime, currentMinute, currentSecond);
    }
}
```

Problem No. 19: Finding large number using switch case

```
//finding large number using switch-case

import java.util.Scanner;

public class Code3_Finding_Large_Num_Sw {

public static void CompareValue(int num1, int num2, int num3,
int num4) {
 switch ((num1 > num2 && num1 > num3 && num1 > num4) ? 1 : 0) {
    case 1:
   System.out.println("1st number: " + num1 + " is the largest
number.");
     break;
        case 0:
   switch ((num2 > num3 && num2 > num4 && num2 > num1) ? 1 : 0)
{
     case 1:
     System.out.println("2nd number: " + num2 + " is the
largest number.");
      break;
     case 0:
  switch ((num3 > num2 && num3 > num4 && num3 > num1) ? 1 : 0)
{
  case 1:
    System.out.println("3rd number: " + num3 + " is the largest
number.");
    break;
      case 0:
   switch ((num4 > num1 && num4 > num2 && num4 > num3) ? 1 : 0)
{
      case 1:
   System.out.println("4th number: " + num4 + " is the largest
number.");
     break;
     case 0:
     System.out.println("They are equal.");
       break;
       }
           break;
         }
             break;
```

```
                }
                break;
        }
    }

    public static void main(String[] args) {
        System.out.print("Enter 4 numbers: ");

        Scanner sc = new Scanner(System.in);

        int num1 = sc.nextInt();
        int num2 = sc.nextInt();
        int num3 = sc.nextInt();
        int num4 = sc.nextInt();

        CompareValue(num1, num2, num3, num4);

    }
}
```

Problem No. 04: Finding large number using if-else.

```
//finding large number using if-else

import java.util.Scanner;

public class Code4_Finding_Large_Num_If {
  public static void CompareValue(int num1, int num2, int num3,
int num4) {
   if (num1 > num2 && num1 > num3 && num1 > num4) {
    System.out.println("1st number: " + num1 + " is the largest
number.");
        } else if (num2 > num1 && num2 > num3 && num2 > num4) {
    System.out.println("2nd number: " + num2 + " is the largest
number.");
        } else if (num3 > num1 && num3 > num2 && num3 > num4) {
    System.out.println("3rd number: " + num3 + " is the largest
number.");
        } else if (num4 > num1 && num4 > num2 && num4 > num3) {
     System.out.println("4th number: " + num4 + " is the
largest number.");
        }
    }

    public static void main(String[] args) {
        System.out.print("Enter 4 numbers: ");

        Scanner sc = new Scanner(System.in);

        int num1 = sc.nextInt();
        int num2 = sc.nextInt();
        int num3 = sc.nextInt();
        int num4 = sc.nextInt();

        CompareValue(num1, num2, num3, num4);
    }
}
```

Problem No. 05: Finding smallest number using if-else.

```
//finding large number using if-else

import java.util.Scanner;

public class Code5_Finding_Smallest_Num_If {
    public static void CompareValue(int num1, int num2, int
num3, int num4) {
        if (num1 < num2 && num1 < num3 && num1 < num4) {
        System.out.println("1st number: " + num1 + " is the
smallest number.");
        } else if (num2 < num1 && num2 < num3 && num2 < num4) {
         System.out.println("2nd number: " + num2 + " is the
smallest number.");
        } else if (num3 < num1 && num3 < num2 && num3 < num4) {
         System.out.println("3rd number: " + num3 + " is the
smallest number.");
        } else if (num4 < num1 && num4 < num2 && num4 < num3) {
         System.out.println("4th number: " + num4 + " is the
smallest number.");
        }
    }

    public static void main(String[] args) {
        System.out.print("Enter 4 numbers: ");

        Scanner sc = new Scanner(System.in);

        int num1 = sc.nextInt();
        int num2 = sc.nextInt();
        int num3 = sc.nextInt();
        int num4 = sc.nextInt();

        CompareValue(num1, num2, num3, num4);
    }
}
```

Problem No. 06: Display time converter

```
//Convert time Second to minutes
import java.util.Scanner;

public class Code6_DisplayTime {
    /**
     * Display time
     */
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        // Prompt the user for input
        System.out.print("Enter an integer for seconds: ");
        int seconds = input.nextInt();
        int minutes = minusHours(seconds);
        int hour = Hour(seconds);
        int remain = Remain(seconds);

  System.out.println(seconds + " seconds is " + hour +" hours
and " + minutes + " minutes & " + remain + " seconds");
    }

    public static int minusHours(int seconds) {
        int minutes = seconds / 60; // Find minutes in seconds
        minutes = minutes % 60;
        return minutes;
    }

    public static int Hour(int seconds) {
        int minutes = seconds / 60;
        int hour = minutes / 60;
        return hour;

    }

    public static int Remain(int seconds) {
        int remain = seconds % 60;
        return remain;
    }

}
```

Problem No. 07: Average of 3 numbers

```
//Average of 3 number
import java.util.Scanner; // Scanner is in the java.util
package
import java.util.function.DoubleConsumer;

public class Code7_Average {
    public static void main(String[] args) {
 System.out.println("Average option\n1.Integer
Average.\n2.Double Avarage.");
        System.out.print("Enter your option: ");
        int a;

        Scanner input = new Scanner(System.in);
        a = input.nextInt();
        switch (a) {
            case 1:
                // Prompt the user to enter three numbers
                System.out.print("Enter three numbers: ");
                int number1 = input.nextInt();
                int number2 = input.nextInt();
                int number3 = input.nextInt();
                // Display result
 System.out.println("The average of " + number1 + " " + number2
     + " " + number3 + " is " +IntegerAverage(number1, number2,
number3));
                break;
            case 2:
                // Prompt the user to enter three numbers
                System.out.print("Enter three numbers: ");
                double x = input.nextDouble();
                double y = input.nextDouble();
                double z = input.nextDouble();
                // Display result
     System.out.println("The average of " + x + " " + y
                  + " " + z + " is " + DoubleAverage(x, y, z));
                break;
            default:

        }

    }
```

```
    public static int IntegerAverage(int number1, int number2,
int number3) {
        int average = (number1 + number2 + number3) / 3; //
Compute average
        return average;
    }

    public static double DoubleAverage(double x, double y,
double z) {
        double average1 = (x + y + z) / 3; // Compute average
        return average1;
    }
}
```

Problem No. 08: Age.java

```
//Age.java
import java.util.Scanner;

public class Code8_Age {
    public static void main(String[] args) {
        final int MINOR = 21;
        Scanner input = new Scanner(System.in);

        System.out.print("Enter you age: ");
        int age = input.nextInt();
        System.out.println("You entered: " + age);
        age(MINOR, age);
    }

    public static void age(int MINOR, int age) {
        if (age < MINOR)
            System.out.println("Youth is a wonderful thing. Enjoy..!!");

        System.out.print("Age is a state of mind.");

    }
}
```

Problem No. 09: Converter Fahrenheit to Celsius

```java
//Temparature Convertor
import java.util.Scanner;

public class Code9_Temparature {
    public static void main(String[] args) {
   System.out.println("Temparature Converter:\n1.Fahrenheit to celsius.\n2.Celsius
to Fahrenheit.");
        System.out.print("Enter your Option: ");
        try (Scanner input = new Scanner(System.in)) {
            int a = input.nextInt();
            switch (a) {
                case 1: {
                    System.out.print("Enter a degree in Fahrenheit: ");
                    double fahrenheit = input.nextDouble();
                    double celsius = Celsius(fahrenheit);
  System.out.println("Fahrenheit " + fahrenheit + " is " + celsius + " in Celsius");
                }
                    break;
                case 2: {
                    System.out.print("Enter a degree in Celsius: ");
                    double celsius = input.nextDouble();
                    double fahrenheit = Fahrenheit(celsius);
  System.out.println("Celsius " + celsius + " is " + fahrenheit + " in fahrenheit");
                }
                    break;
                default:

            }
        }
    }

    public static double Celsius(double fahrenheit) {
        // Convert Fahrenheit to Celsius
        double celsius = (5.0 / 9) * (fahrenheit - 32);
        return celsius;
    }

    public static double Fahrenheit(double celsius) {
        // Convert Celsius to Fahrenheit
        double fahrenheit = ((1.8 * celsius) + 32);
        return fahrenheit;
    }
}
```

Problem No. 10: Shapes Area

```
//Shapes Area
import java.util.Scanner;

public class Code10_ShapesArea {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.println("\n\tArea Calculator");
     System.out.println("1. Area of Circle\n2. Area of Cube\n3. Area
of Rectengle\n4. Area of Square\n5. Area of Tringle\n6. Area of
Trapezium.");
        System.err.print("Select One of them:");
        int shape = input.nextInt();
        double length, area;
        switch (shape) {
            case 1:
        System.out.print("\tArea of Circle.\nEnter the radius:");
                double r = input.nextDouble();
                area = CircleArea(r);
                System.out.println("Area of Circle is: " + area);
                break;
            case 2:
                double side, volume, surfaceArea;

System.out.print("\tArea of Cube.\nEnter Length of Side of Cube: ");
                side = input.nextDouble();

                surfaceArea = cubeArea(side);
                /* Volume of Cube = side X side X side */
                volume = side * side * side;

   System.out.println("Surface Area of Cube =" + surfaceArea);
                System.out.print("Volume of Cube =" + volume);
                break;
            case 3:
   System.out.print("\tArea of Rectangle\nEnter the length of
Rectangle:");
                length = input.nextDouble();
                System.out.print("Enter the width of Rectangle:");
                double width = input.nextDouble();
                area = RactangleArea(length, width);
                System.out.println("Area of Rectangle is:" + area);
                break;
            case 4:
```

```
  System.out.print("\tArea of Square\nEnter the length of Squre:");
                length = input.nextDouble();
                area = SqureArea(length);
                System.out.println("Area of the Squre is:" + area);
                break;
            case 5:
  System.out.print("\tArea of Triangle\nEnter the width of the
Triangle:");
                double b = input.nextDouble();

  System.out.print("Enter the height of the Triangle:");
                double h = input.nextDouble();

                area = TriangleArea(b, h);
                System.out.println("Area of Triangle is: " + area);
                break;
            case 6:
   System.out.print("Enter the length of the parallel sides of the
trapezium: ");
                double side1 = input.nextDouble(); // Take input
from the user
   System.out.print("Enter the length of the parallel sides of the
trapezium: ");
                double side2 = input.nextDouble();
   System.out.print("Enter the height of the trapezium: ");
                double height = input.nextDouble();
                area = trapeziumArea(side1, side2, height);
   System.out.println("Area of the Trapezium is: " + area);
                break;
        }
    }

    private static double cubeArea(double side) {
        /* Total surface area of Cube = 6 X side X side */
        double surfaceArea = 6 * side * side;
        return surfaceArea;
    }

    public static double CircleArea(double r) {
        double area = (22 * r * r) / 7;
        return area;
    }

    public static double RactangleArea(double length, double width)
{
```

```java
        // Area = length*width;
        double area = length * width;
        return area;
    }

    public static double SqureArea(double length) {
        double area = length * length;
        return area;
    }

    public static double TriangleArea(double b, double h) {
        double area = (b * h) / 2; // Area = (width*height)/2
        return area;
    }

    public static double trapeziumArea(double side1, double side2,
double height) {
 double area = ((side1 + side2) * height) / 2; // Calculate the area
        if (side1 <= 0 || side2 <= 0)
            System.out.println("Length should be positve");
        else
            System.out.println("Area of trapezium = " + area);
        return area;
    }
}
```

Problem No. 11: Wages.java

```
//Wages.java
import java.text.NumberFormat;
import java.util.Scanner;

public class Code11_Wages {
    public static void main(String[] args) {
        final double RATE = 8.25;
        final int STANDARD = 40;
        try (Scanner scan = new Scanner(System.in)) {
            // double pay = 0.0;

   System.out.print("Enter the number of hours worked: ");
            int hours = scan.nextInt();
            System.out.println("");
            double pay = Pay(hours, RATE, STANDARD);
     NumberFormat fmt = NumberFormat.getCurrencyInstance();
      System.out.println("Gross earning: " + fmt.format(pay));

        }
    }

    public static double Pay(int hours, double RATE, int
STANDARD) {
        double pay;
        if (hours > STANDARD)
            pay = STANDARD * RATE + (hours - STANDARD) * (RATE
* 1.5);
        else
            pay = hours * RATE;
        return pay;
    }

}
```

Problem No. 12: Guessing number.

```
//Number Guessing
import java.util.*;

public class Code12_NumberGuessing {
    public static void main(String[] args) {
        final int MAX = 10;
        Scanner scan = new Scanner(System.in);
        Random generator = new Random();
        int answer = generator.nextInt(MAX) + 1;
 System.out.println("I'm thinking of a number between 1 & and "
+ MAX + ".");
        System.out.print("Guess what it is: ");
        int guess = scan.nextInt();
        Guess(guess, answer);
    }

    public static void Guess(int guess, int answer) {
        if (guess == answer) {
            System.out.println("You got it! Good guessing!");
        } else {
            System.out.println("That is not correct!");
            System.out.println("The number was " + answer);
        }
    }
}
```

Problem No. 13: Quadratic Equation calculator.

```java
//Quadratic.java
import java.util.Scanner;

public class Code13_Quadratic {
    public static void main(String[] args) {
        // double root1 = 0.0, root2 = 0.0;
        Scanner input = new Scanner(System.in);
        System.out.println("\n\n\tSolving The Equation ax^2+bx+c=0");
        System.out.print("Enter (a) the coefficient of squared x: ");
        double a = input.nextDouble();
        System.out.print("Enter (b) the coefficient of x: ");
        double b = input.nextDouble();
        System.out.print("Enter (c) the Constant: ");
        double c = input.nextDouble();
        double d = discriminant(a, b, c);

        if (d > 0.0) {
            double root1 = ((-1 * b + Math.sqrt(d)) / 2 * a);
            double root2 = ((-1 * b - Math.sqrt(d)) / 2 * a);
            System.out.println("\n\t#Root 1:: " + root1 + "\n\t#Root 2:: " + root2);
        } else if (d == 0.0) {
            double root = (-b / 2.0 * a);
            System.out.println("\n\t#Root :: " + root);
        } else {
            System.out.println("\n\tRoots are not real. They are Complex number");
        }
    }

    public static double discriminant(double a, double b, double c) {
        double discriminant = Math.pow(b, 2) - (4 * a * c);
        return discriminant;
    }

    /*
     * public static double Root1(double a, double b, double d, double c) {
     * double root1 = ((-1 * b + Math.sqrt(d)) / 2 * a);
     * return root1;
     * }
     *
     * public static double Root2(double a, double b, double d, double c) {
     * double root2 = ((-1 * b - Math.sqrt(d)) / 2 * a);
     * return root2;
     *
     * }
     */

}
```

Problem No. 14: Random characters.

```
//random character.java
import java.util.Scanner;

public class Code14_RandomCharacter {
    public static void main(String[] args) {
    System.out.println("\n\nEnter The range of
character:");
        Scanner sc = new Scanner(System.in);
        char ch1 = sc.next().charAt(0);
        char ch2 = sc.next().charAt(0);
        char r = RandomChar(ch1, ch2);
        System.out.print(r);

    }

public static char RandomChar(char ch1, char ch2)
{
        System.out.print("A Random Character
between " + ch1 + " & " + ch2 + " is ");
        char r = (char) (ch1 + Math.random() *
(ch2 - ch1 + 1));
        return r;
    }
}
```

Problem No. 15: Class character.

```java
//Class character test
import java.util.Scanner;

public class Code15_ClassCharacterTest {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter any two Characters: ");
        Character ch1 = input.next().charAt(0);
        Character ch2 = input.next().charAt(0);
        testing1(ch1, ch2);

        char ch3 = 'R';
        char ch4 = '7';
        char ch5 = '*';

        testing2(ch3, ch4, ch5);
    }

    public static void testing1(char ch1, char ch2) {
        System.out.println(Character.isLowerCase(ch1));
        System.out.println(Character.isLetterOrDigit(ch1));
        System.out.println(Character.isDigit(ch1));
        System.out.println(Character.isDigit(ch2));
        System.out.println(Character.toUpperCase(ch1));
    }

    public static void testing2(char ch3, char ch4, char ch5) {
        System.out.println(Character.isLowerCase(ch3));
        System.out.println(Character.isLetterOrDigit(ch5));
        System.out.println(Character.isDigit(ch4));
        System.out.println(Character.isDigit(ch5));
        System.out.println(Character.isLowerCase(ch3));
   }
}
```

Problem No. 16: Comparing character type.

```
//Compareing Character Types
import java.util.Scanner;

public class Code16_CompareingCharType {
    public static void main(String[] args) {
System.out.println("Comparing Character Type.\nUpper
case or Lower case");
        System.out.print("Enter Any Character: ");
        Scanner sc = new Scanner(System.in);
        char ch = sc.next().charAt(0);
        Chartest(ch);
    }

    public static void Chartest(char ch) {

        if (ch >= 'A' && ch <= 'Z')
   System.out.println(ch + " is an uppercase letter");
        else if (ch >= 'a' && ch <= 'z')
   System.out.println(ch + " is a lowercase letter");
        else if (ch >= '0' && ch <= '9')
   System.out.println(ch + " is a numeric Character");

    }
}
```

Problem No. 17: String Mutation.

```
//String Mutation.java
public class Code17_StringMutation {
    public static void main(String[] args) {
        String pharse = "Change is inevitable";
System.out.println("Orginal String: \"" +pharse +
"\"");
       System.out.println("Length of String:
"+pharse.length());

        Mutation(pharse);
    }

    public static void Mutation(String pharse) {
String mutation1 = pharse.concat(", expect from vending
machines");
        String mutation2 = mutation1.toUpperCase();
        String mutation3 = mutation2.replace('E', 'X');
        String mutation4 = mutation3.substring(3, 30);

   System.out.println("Mutation #1: " + mutation1);
   System.out.println("Mutation #2: " + mutation2);
   System.out.println("Mutation #3: " + mutation3);
   System.out.println("Mutation #4: " + mutation4);

System.out.println("Mutated length:"+mutation4.length()
);
    }
}
```

Problem No. 18: String Concatenation.

```
//String Concatenation
public class Code18_StringConcatenation {
    public static void main(String[] args) {
        // three String are Concatenated
     String message = "Welcome" + " to" + " java.";

     // String Chapter is Concatenated with number 2
     String s = "Chapter " + 2; // s become chapter 2

  // string Supplement is concatenated with character B
    String s1 = "Supplement" + 'B'; // s1 becomes
SupplementB;

       printer(message, s, s1);
    }

public static void printer(String message, String s,
String s1) {
        System.out.println(message);
        System.out.println(s);
        System.out.println(s1);
    }
}
```

Problem No. 19: String Method.

```
//string Methods
public class Code19_StringMethods {
    public static void main(String[] args) {
        String s1 = "Welcome";
        // String s2 = new String(char[]);
        String s2 = "    Hello!      ";
        char ch = s1.charAt(0);
        int length = s1.length();
        int index = s1.indexOf(ch);
        int inx = s1.lastIndexOf(ch);
        boolean b = s1.equals(s2);
        boolean c = s1.equalsIgnoreCase(s2);
        boolean d = s1.startsWith(s2);
        boolean e = s1.endsWith(s2);

        String s = s1.toUpperCase();
        String t = s1.trim();

        printer(s1, s2, s2, index, inx, b, c, d, e, s, t);
    }

public static void printer(String s1, String s2, String ch, int
index, int inx, boolean b, boolean c, boolean d,
            boolean e, String s, String t)
{
        System.out.println(s1);
        System.out.println(s2);
        System.out.println(ch);
        System.out.println(index);
        System.out.println(inx);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
        System.out.println(e);
        System.out.println(s);
        System.out.println(t);
    }
}
```

Problem No. 20: Comparing String

```
//Comparing string type
import java.util.Scanner;

public class Code20_ComparingString {
    public static void main(String[] args) {

        Scanner in = new Scanner(System.in);

        System.out.print("Enter any Two String:");
        String s1 = in.next();
        String s2 = in.next();

        Compare(s1, s2);
    }

    public static void Compare(String s1, String s2) {
        boolean b = s1.equals(s2);
        boolean c = s1.equalsIgnoreCase(s2);
        boolean d = s1.startsWith(s2);
        boolean e = s1.endsWith(s2);
        int a = s1.compareTo(s2);
        int f = s1.compareToIgnoreCase(s2);
        boolean g = s1.startsWith(s2);
        boolean h = s1.endsWith(s2);

        System.out.println(a + "\n" + b + "\n" + c +
"\n" + d + "\n" + e + "\n" + f + "\n" + g + "\n" + h);
    }
}
```

Problem No. 21: Conversion of String

```
//Conversion of String
public class Code21_Conversion_of_string {
    public static void main(String[] args) {
        String intString = "123";
   int intValue = Integer.parseInt(intString);

        String doubleString = "123.456";
    double doubleValue =
Double.parseDouble(doubleString);

        int number = 324432;
        String s = "" + number; // gives "324432"

        System.out.println(intString);
        System.out.println(intValue);
        System.out.println(doubleString);
        System.out.println(doubleValue);
        System.out.println(number);
        System.out.println(s);
    }
}
```

Problem No. 22: Using All Math Classes & Methods

```java
//Math Class Methods
import java.util.Scanner;

public class Code22_MathClassMethod {
    public static void main(String[] args) {
        System.out.println("Value of PI " + Math.PI);
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Any degre for Trigonomatric Function:");
        double a = sc.nextDouble();
        Trigonomatric(a);

        System.out.print("Enter Any value for Exponent:");
        double b = sc.nextDouble();
        Exponent(b);
    }

    public static double Trigonomatric(double a) {
        System.out.println("Value of Sin " + a + " is " + Math.sin(a));
        System.out.println("Value of Cos " + a + " is " + Math.cos(a));
        System.out.println("Value of Tan " + a + " is " + Math.tan(a));
 System.out.println("Value of " + a + "Cos" + a + " is " + Math.acos(a));
 System.out.println("Value of " + a + "Tan" + a + " is " + Math.atan(a));
 System.out.println("Value of " + a + "Sin" + a + " is " + Math.asin(a));
        return a;
    }

    public static double Exponent(double b) {
 System.out.println("Value of exponent " + b + " is " + Math.exp(b));
 System.out.println("Value of log " + b + " is " + Math.log(b));
 System.out.println("Value of log10 " + b + " is " + Math.log10(b));
 System.out.println("Value of " + b + " Square is " + Math.pow(b, 2));
 System.out.println("Value of Square Root " + b + " is " + Math.sqrt(b));
        return b;
    }
}
```

Problem No. 23: Sentinel Value Summation.

```
//Sentinel Value Summation
import java.util.Scanner;

public class Code23_SentanelValueSummation {

    public static void main(String[] args) {
        int value;
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an Integer (0 To Quit) : ");
        value = sc.nextInt();

        SentanelValue(value);
    }

    public static void SentanelValue(int value) {
        int sum = 0, count = 0;
        while (value != 0) {
            count = count + 1;
            sum = sum + value;
       System.out.println("The sum is so far is " + sum);
       System.out.println("Enter the Integer (0 To Quit) : ");
            Scanner s = new Scanner(System.in);
            value = s.nextInt();
        }
        if (count == 0) {
            System.out.println("No Values were enterd");
        } else {
            System.out.println("Sum of all values= " + sum);
        }

    }
}
```

Problem No. 24: Win percentage

```
//Win Percentage
import java.text.NumberFormat;
import java.util.Scanner;

public class Code24_WinPercentage {
    public static void main(String[] args) {
        final int NUM_GAMES = 12;
        int won;
        double ratio = 0;
        Scanner sc = new Scanner(System.in);
System.out.print("Enter the number of games won (0 to" +
NUM_GAMES + "): ");
        won = sc.nextInt();
        Percentage(won, ratio, NUM_GAMES);
    }

    public static void Percentage(int won, double ratio, int
NUM_GAMES) {
        Scanner sc = new Scanner(System.in);
        while (won < 0 || won > NUM_GAMES) {
System.out.print("Invalid input.\nPlease reenter:");
            won = sc.nextInt();
        }
        ratio = (double) won / NUM_GAMES;
        NumberFormat fmt = NumberFormat.getPercentInstance();
        System.out.println();
System.out.println("Winning percentage: " + fmt.format(ratio));
    }
}
```

Problem No. 25: Reverse number.

```java
//reverse number
import java.util.Scanner;

public class Code25_ReverseNumber {
    public static void main(String[] args) {
        int number, reverse = 0;
        Scanner sc = new Scanner(System.in);
  System.out.print("\n\nEnter a positive integer: ");
        number = sc.nextInt();
        reverse = Reverse(number);
  System.out.println("That number reverse is: " +
reverse);
    }

    public static int Reverse(int number) {
        int reverse = 0;
        do {
            int lastDigit = number % 10;
            reverse = (reverse * 10) + lastDigit;
            number = number / 10;
        } while (number > 0);
        return reverse;

    }
}
```

Problem No. 26: Multiples

```
// MUltiples Numbers
import java.util.Scanner;

import javax.swing.plaf.multi.MultiButtonUI;

public class Code26_Multiples {
    public static void main(String[] args) {
        final int PER_LINE = 5;
        int value, limit;
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a posivite value : ");
        value = sc.nextInt();
        System.out.println("Enter an upper limit : ");
        limit = sc.nextInt();
System.out.println("\nThe multiple of " + value + " between " +
value + " & " + limit + " (inclusive) are :");
        Multiples(value, limit, PER_LINE);

    }

public static void Multiples(int value, int limit, int
PER_LINE) {
        int multiple = 0, count = 0;
for (multiple = value; multiple <= limit; multiple = multiple +
value) {
            System.out.print(multiple + "\t");

            count = count + 1;
            if (count % PER_LINE == 0) {
                System.out.println("");
            }
        }
    }
}
```

Problem No. 27: Example of infinity loops

```
//Infinity loop example
import java.util.Scanner;

public class Code27_InfinityLoop {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
System.out.print("Tomorrow is your Exam\ntrue or
false\n\nans:");
        boolean b = sc.nextBoolean();
        Loop(b);
    }

    public static void Loop(boolean b) {
        if (b == true) {
            int i, j = 0;
            for (i = 0; i <= j; i--) {
           System.out.print("Go to study!!\t");
            }
        } else
System.out.println("Take a coffee & chill.\nLife is
Good");
    }
}
```

Problem No. 28: Palindrome Tester

```
//Palindrome Testing
import java.util.Scanner;
import javax.swing.plaf.synth.SynthSplitPaneUI;

public class Code28_PalindromeTester {
    public static void Palindrome(String another) {
        int left, right;
        Scanner sc = new Scanner(System.in);
        while (another.equalsIgnoreCase("y")) {
   System.out.print("Enter a potential palidrome string : ");
            String str = sc.nextLine();
            left = 0;
            right = str.length() - 1;
            while (str.charAt(left) == str.charAt(right) &&
left < right) {
                left = left + 1;
                right = right - 1;
            }
            System.out.println();
            if (left < right) {
   System.out.println("The String are not a Palindrome");
            } else
   System.out.println("That String is Palindrome.\n");

   System.out.println("Test palindrome again..??");
            another = sc.nextLine();
        }
    }

    public static void main(String[] args) {
        String another = "y";
        Palindrome(another);
    }
}
```

Problem No. 29: Triangle Shape star pattern

```java
//Triangle star Pattern
import java.util.Scanner;

public class Code29_TrinaglePattern {
    public static void main(String[] args) {
        int i = 1, j = 1;
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the line number: ");
        int n = sc.nextInt();
        Triangle(n, i, j);
        InverseTriangle(n, i, j);
    }

    public static void Triangle(int n, int i, int j) {
        for (i = 1; i <= n; i++) {
            for (j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.print("\n");
        }
    }

    public static void InverseTriangle(int n, int i, int j) {
        for (i = n - 1; i >= 0; i--) {
            for (j = 0; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.print("\n");
        }
    }
}
```

Problem No. 30: Sum method.

```
//Sum Method
public class Code30_SumMethod {
    public static void main(String[] args) {
        int result = sum(1, 10);
        System.out.println("Sum from 1 to 10 is:\t" + result);
    }

    public static int sum(int num1, int num2) {
        int sum = 0;
        for (int i = num1; i <= num2; i++) {
            sum = sum + i;
        }
        return sum;
    }
}
```

Problem No. 31: Test Max value

```
// max number function testing
public class Code31_TestMax {
    public static void main(String[] args) {
        int i = 5;
        int j = 2;
        int k = max(i, j);
  System.out.println("The maximum of " + i + " & " + j + " is "
+ k);
    }

    public static int max(int num1, int num2) {
        int result;
        if (num1 > num2) {
            result = num1;
        } else
            result = num2;

        return result;
    }
}
```

Problem No. 32: Test Method

```
public class Code32_TestMethod {
    public static void main(String[] args) {
        int a =10;
        int b= 20;
        int addResult = Add(a,b);
        System.out.println("Sum of a & b is : "+addResult);
    }
    public static int Add(int x, int y) {
        return (x+y);
    }
}
```

Problem No. 33: Test pass by value

```
public class Code33_TestPassByValue {
    public static void main(String[] args) {
        int num1 =1;
        int num2 =2;
 System.out.println("Before calling swap:\nNum1 = "+num1+" Num2
= "+num2+"\n");
        swap(num1,num2);
 System.out.println("After calling swap:\nNum1 = "+num1+" Num2
= "+num2+"\n");
    }
    public static void swap(int n1, int n2){
 System.out.println("Inside swap, before Swapping:\nn1 = "+n1+"
n2 = "+n2+"\n");
        int temp = n1;
        n1 =n2;
        n2 =temp;
 System.out.println("Inside swap, after swapping:\nn1 = "+n1+"
n2 = "+n2+"\n");
    }}
```

Problem No. 34: Ambiguous Over Loading Error

```
//Ambiguous Over Loading Error
public class Code34_AmbiguousOverLoading {
    public static void main(String[] args) {
        System.out.println(Max(1,2));
    }
    public static double Max(int num1, double num2) {
        if(num1>num2)
        return num1;
        else
        return num2;

    }
    public static double Max(double num1, int num2) {
        if (num1>num2){
            return num1;
        }
        else
        return num2;

    }
}
```

Problem No. 35: Factorial using loop.

```
import java.util.Scanner;

//Factorial using Loop
public class Code35_FactorialLoop {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your number for factorial!: ");
    int number = sc.nextInt();// It is the number to calculate
factorial

        Factorial(number);
    }

    public static int Factorial(int number) {
        int fact = 1;
        for (int i = 1; i <= number; i++) {
            fact = fact * i;
        }
  System.out.println("Factorial " + number + "! is: " + fact);
        return fact;
    }
}
```

Problem No. 36: Factorial using recursion.

```
import java.util.Scanner;

// Factorial Using Recursion
public class Code36_FactorialRecursion {
    public static int factorial(int n) {
        if (n == 0)
            return 1;
        else
            return (n * factorial(n - 1));
    }

    public static void main(String args[]) {
        int i, fact = 1;
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your number for factorial!: ");
      int number = sc.nextInt();// It is the number to
calculate factorial
        fact = factorial(number);
  System.out.println("Factorial " + number + "! is: " + fact);
    }
}
```

Problem No. 37: Fibonacci Number

```
// Fibonacci Series Using Recursion
public class Code37_FibonacciRecursion {
    static int n1 = 0, n2 = 1, n3 = 0;

    static void printFibonacci(int count) {
        if (count > 0) {
            n3 = n1 + n2;
            n1 = n2;
            n2 = n3;
            System.out.print(" " + n3);
            printFibonacci(count - 1);
        }
    }

    public static void main(String args[]) {
        int count = 10;
        System.out.print("The Fibonacci Series (Using Recursion) : ");
        System.out.print(n1 + " " + n2);// printing 0 and 1
        printFibonacci(count - 2);// n-2 because 2 numbers are already
printed
    }
}
```

Problem No. 38: Box Character Pattern.

```
//Box Pattern Character
public class Code38_BoxCharacter {

    public static void main(String[] args) {
        char c=65;
        int i=1;
        Pattern(c,i);
    }
    public static void Pattern(char c, int i) {
        for (c = 65; c <= 69; c++) {
            System.out.print(c + " ");
            if (c == 65 || c == 69) {
                for (i = 1; i <= 5; i++) {
                    System.out.print(c + " ");
                }
            } else {
                for (i = 1; i <= 5; i++) {
                    System.out.print("  ");
                }
            }

            System.out.println(c);
        }
    }
}
```